

3.1 Data Type Correction Table — All 18 Columns

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fx

= Table.TransformColumnTypes(#"Promoted Headers",{{"Order_ID", type text}, {"Customer_ID", type text}, {"Date", type date}, {"Age", Int64.Type}, {"Gender", type text}, {"City", type text}, {"Product_Category", type text}, {"Unit_Price", type number}, {"Quantity", Int64.Type}, {"Discount_Amount", type number}, {"Total_Amount", type number}, {"Payment_Method", type text}, {"Device_Type", type text}, {"Session_Duration_Minutes", type number}, {"Pages_Viewed", Int64.Type}, {"Is_Returning_Customer", type logical}, {"Delivery_Time_Days", Int64.Type}, {"Customer_Rating", Int64.Type}})

	A ^B Order_ID	A ^B Customer_ID	A ^B Date	1 ² Age	A ^B Gender	A ^B City	A ^B Product
1	ORD_001337	CUST_01337	1/1/2023		27 Female	Bursa	Toys
2	ORD_004885	CUST_04885	1/1/2023		42 Male	Konya	Toys
3	ORD_004507	CUST_04507	1/1/2023		43 Female	Ankara	Food
4	ORD_000645	CUST_00645	1/1/2023		32 Male	Istanbul	Electronics
5	ORD_000690	CUST_00690	1/1/2023		40 Female	Istanbul	Sports
6	ORD_000223	CUST_00223	1/1/2023		43 Female	Istanbul	Beauty
7	ORD_002506	CUST_02506	1/1/2023		25 Female	Izmir	Electronics
8	ORD_004540	CUST_04540	1/1/2023		44 Male	Konya	Food
9	ORD_001808	CUST_01808	1/2/2023		41 Male	Istanbul	Fashion
10	ORD_003413	CUST_03413	1/2/2023		58 Male	Istanbul	Sports
11	ORD_004457	CUST_04457	1/2/2023		21 Female	Istanbul	Electronics
12	ORD_004797	CUST_04797	1/2/2023		40 Male	Konya	Beauty
13	ORD_001211	CUST_01211	1/2/2023		20 Female	Bursa	Sports
14	ORD_000514	CUST_00514	1/2/2023		24 Male	Eskisehir	Home & Gard
15	ORD_003462	CUST_03462	1/2/2023		57 Female	Istanbul	Books
16	ORD_001418	CUST_01418	1/2/2023		38 Male	Istanbul	Food
17	ORD_000198	CUST_00198	1/2/2023		36 Male	Istanbul	Books
18	ORD_000374	CUST_00374	1/2/2023		46 Female	Bursa	Sports
19	ORD_003204	CUST_03204	1/3/2023		50 Female	Ankara	Sports
20	ORD_004204	CUST_04204	1/3/2023		23 Male	Bursa	Electronics
21	ORD_000751	CUST_00751	1/3/2023		23 Female	Antalya	Food
22	ORD_002466	CUST_02466	1/3/2023		45 Male	Istanbul	Electronics
23	ORD_003920	CUST_03920	1/3/2023		34 Male	Kayseri	Fashion
24	ORD_000038	CUST_00038	1/4/2023		18 Female	Izmir	Beauty
25	ORD_003721	CUST_03721	1/4/2023		46 Male	Istanbul	Electronics

The DATA_QUALITY sheet provides a detailed record of the data integrity checks performed across all 18 columns. It captures key dimensions such as null counts, percentages of missing values, duplicate checks, and outlier detection using the IQR method, along with the corrective actions taken for each issue.

	A	B	C	D	E	F	G	H
1								
2	Column Name	Null Count	% Missing	Full Duplicates	Duplicate Order	Type Fix Applied	Outliers Flag	Action Taken
3	Order_ID	0	0%	0	0	YES	N/A	NONE
4	Customer_ID	0	0%	0	0	YES	N/A	NONE
5	Date	0	0%	0	0	YES	N/A	NONE
6	Age	0	0%	0	0	YES	N/A	NONE
7	Gender	0	0%	0	0	VERIFIED	N/A	NONE
8	City	0	0%	0	0	VERIFIED	N/A	NONE
9	Product_Category	0	0%	0	0	TRIMMED	N/A	NONE
10	Unit_Price	0	0%	0	0	VERIFIED	513 ROWS	CAPPED
11	Quantity	0	0%	0	0	VERIFIED	0	NONE
12	Discount_Amount	0	0%	0	0	VERIFIED	N/A	NONE
13	Total_Amount	0	0%	0	0	VERIFIED	546 ROWS	CAPPED
14	Payment_Method	0	0%	0	0	VERIFIED	N/A	NONE
15	Device_Type	0	0%	0	0	VERIFIED	N/A	NONE
16	Session_Duration_Minutes	0	0%	0	0	VERIFIED	124 ROWS	KEPT
17	Pages_Viewed	0	0%	0	0	VERIFIED	N/A	NONE
18	Is_Returning_Customer	0	0%	0	0	YES	N/A	NONE
19	Delivery_Time_Days	0	0%	0	0	VERIFIED	141 ROWS	KEPT
20	Customer_Rating	0	0%	0	0	VERIFIED	N/A	NONE
21								
22	Column	Q1	Q3	IQR	UPPERFENCE	OUTLIERCOUNT	ACTIONTAKE	
23	Unit_Price	76,5875	513,93	437,3425	1169,94375	513	CAP AT UPPER	
24	Quantity	1	3	2	6	0	NO ACTION	
25	Total_Amount	122,5175	979,695	857,1775	2265,46125	546	CAPP UPPER	
26	Session_Duration_Minutes	8	19	11	35,5	124	KEEP DOCUMENT	
27	Delivery_Time_Days	4	8	4	14	141	KEEP DOCUMENT	

The dataset is in excellent condition, with no null values and no duplicate records across any of the 18 columns. The only data quality interventions required involved managing outliers in *Unit Price*, *Total_Amount*, *Session_Duration_Minutes*, and *Delivery_Time_Days*. These were addressed using the IQR method to ensure consistency and reliability in the analysis.

3.3 Total_Amount Validation

Total_Amount was validated against the formula: $\text{Total_Amount} = (\text{Unit_Price} \times \text{Quantity}) - \text{Discount_Amount}$. A calculated column [Calc_Total] was created in Power Query and compared against the stored Total_Amount. Result: 0 mismatches out of 5,000 rows.

3.4 Feature Engineering — 9 New Columns

Nine additional columns were introduced in Power Query and as DAX measures to enhance the dataset. These support advanced time-series analysis, customer segmentation, and behavioural profiling. For transparency, each column's definition, calculation formula, and the business rationale behind its inclusion are documented below.

S	T	U	V	W	X	Y	Z	AA	AB
Order_Year	Order_Quarter	Order_MonthName	Day_Of_Week	Discount_Flag	Age_Group	Discount_Rate	Engagement_Tier	Page_Bins	Delivery_Speed
2023 Q1	January	January	Sunday	FALSE	18-50		0 Medium	10+	Slow (8+ days)
2023 Q1	January	January	Sunday	FALSE	18-50		0 Medium	1-3	Fast (1-3 days)
2023 Q1	January	January	Sunday	FALSE	18-50		0 Medium	7-9	Standard (4-7 days)
2023 Q1	January	January	Sunday	TRUE	18-50	28,51528493	Medium	10+	Fast (1-3 days)
2023 Q1	January	January	Sunday	FALSE	18-50		0 High	10+	Standard (4-7 days)
2023 Q1	January	January	Sunday	FALSE	18-50		0 Medium	7-9	Slow (8+ days)
2023 Q1	January	January	Sunday	FALSE	18-50		0 Medium	4-6	Standard (4-7 days)
2023 Q1	January	January	Sunday	FALSE	18-50		0 Medium	10+	Fast (1-3 days)
2023 Q1	January	January	Monday	TRUE	18-50	4,824935952	Medium	7-9	Fast (1-3 days)
2023 Q1	January	January	Monday	TRUE	51+	6,867208292	Medium	4-6	Standard (4-7 days)
2023 Q1	January	January	Monday	FALSE	18-50		0 Medium	4-6	Fast (1-3 days)
2023 Q1	January	January	Monday	TRUE	18-50	9,875421749	Medium	4-6	Standard (4-7 days)
2023 Q1	January	January	Monday	FALSE	18-50		0 Medium	7-9	Fast (1-3 days)
2023 Q1	January	January	Monday	FALSE	18-50		0 Medium	7-9	Slow (8+ days)
2023 Q1	January	January	Monday	TRUE	51+	5,888679753	Medium	7-9	Standard (4-7 days)
2023 Q1	January	January	Monday	FALSE	18-50		0 Medium	7-9	Standard (4-7 days)
2023 Q1	January	January	Monday	FALSE	18-50		0 Medium	10+	Standard (4-7 days)
2023 Q1	January	January	Monday	FALSE	18-50		0 High	10+	Standard (4-7 days)
2023 Q1	January	January	Tuesday	TRUE	18-50	14,38204827	High	10+	Slow (8+ days)
2023 Q1	January	January	Tuesday	TRUE	18-50	4,253622912	High	7-9	Slow (8+ days)
2023 Q1	January	January	Tuesday	TRUE	18-50	7,287985866	Medium	10+	Fast (1-3 days)
2023 Q1	January	January	Tuesday	TRUE	18-50	20,42171069	Medium	10+	Fast (1-3 days)
2023 Q1	January	January	Tuesday	FALSE	18-50		0 Medium	4-6	Standard (4-7 days)
2023 Q1	January	January	Wednesday	FALSE	18-50		0 Medium	7-9	Slow (8+ days)
2023 Q1	January	January	Wednesday	TRUE	18-50	5,500147451	Medium	7-9	Slow (8+ days)
2023 Q1	January	January	Wednesday	TRUE	51+	7,676339934	High	10+	Slow (8+ days)
2023 Q1	January	January	Wednesday	FALSE	18-50		0 Medium	7-9	Standard (4-7 days)
2023 Q1	January	January	Wednesday	TRUE	51+	1,745212177	Medium	10+	Standard (4-7 days)
2023 Q1	January	January	Wednesday	FALSE	18-50		0 Medium	4-6	Slow (8+ days)
2023 Q1	January	January	Wednesday	TRUE	18-50	4,51237906	High	10+	Fast (1-3 days)
2023 Q1	January	January	Wednesday	FALSE	18-50		0 Medium	4-6	Slow (8+ days)
2023 Q1	January	January	Wednesday	TRUE	18-50	5,147596638	High	10+	Slow (8+ days)

Power Query transformations were applied in a structured sequence to ensure data integrity and readiness for analysis:

1. Load raw data
2. Promote headers
3. Set data types
4. Trim text columns
5. Validate *Total_Amount*
6. Add nine engineered columns
7. Cap outliers for *Unit_Price* and *Total_Amount*

The engineered columns form the foundation for all pivot table analyses presented in Section 4.

3.5 RFM Methodology

RFM (Recency, Frequency, Monetary) analysis is a proven customer segmentation framework that ranks customers across three behavioural dimensions:

- **Recency** — how recently a customer purchased
- **Frequency** — how often they purchase
- **Monetary** — how much they spend

By combining these dimensions, businesses can identify high-value segments (e.g., “Champions” or “Loyal Customers”) and tailor retention strategies accordingly. This approach ensures marketing investment is directed toward the customers most likely to deliver long-term value, while also highlighting at-risk segments that require re-engagement.

Business Implication

Applying RFM analysis enables data-driven retention strategies such as loyalty programmes, personalised offers, and targeted win-back campaigns. It shifts focus from broad acquisition to precision retention, maximising ROI by nurturing the customers who matter most.

RFM Scoring Approach: Each dimension is scored 1-5 using quintile ranking (5 = best). A composite RFM score is computed as $R_Score + F_Score + M_Score$ (range 3-15). Customers scoring 12-15 are classified as Champions, 9-11 as Loyal Customers, 6-8 as Potential Loyalists, and 3-5 as At-Risk customers. This segmentation directly informs marketing spend allocation and retention programme design.

Section 4: Descriptive Analysis

All findings are derived from PivotTable analyses of the cleaned 5,000-row dataset. The sub-sections below correspond to Tasks 2.1 through 2.5.

4.1 Sales Performance Overview (Task 2.1)

The KPIs in this dashboard provide a snapshot of the company's financial health, customer behaviour, and operational performance, with several clear business implications

A	B
KPI	VALUES
TOTAL REVENUE	4915544,57
TOTAL ORDERS	5000
AOV	983,108914
AVG RATING	3,9028
AVERAGE DELIVERY TIME	6,497
DISCOUNT PENETRATION RATE	30,52
RETURNING VS NEW CUSTOMER SPLIT	59,8
TOP PRODUCT CATEGORY REVENUE	Electronics
TOP CITY BY REVENUE	Istanbul

Business Implication

The high total revenue and strong average order value suggest that customers are spending significantly, which is positive, but the reliance on discounts (over 30% penetration) raises concerns about margin erosion and customer dependence on promotions. The customer split shows nearly 60% are returning buyers, indicating loyalty and repeat business, yet growth may be constrained if new customer acquisition is not strengthened. Operationally, the average delivery time of 6.5 days is relatively slow and could undermine competitiveness, especially in markets where faster fulfillment is the norm. Customer satisfaction, reflected in an average rating of 3.9, is adequate but leaves room for improvement, and enhancing product quality or service could push ratings higher and reduce churn. Market insights reveal that revenue is concentrated in electronics and geographically in Istanbul, which highlights both strengths and risks—dependence on one category and one city makes the business vulnerable to shifts in demand or local market

conditions. Overall, while the company demonstrates strong revenue and loyalty, management should focus on improving delivery speed, reducing reliance on discounts, diversifying product categories and regions, and enhancing customer experience to sustain long-term growth and competitiveness.

4.2 Sales Trends & Seasonality (Task 2.2)

Monthly Revenue — 2023 Full Year

A	B	C	D	E	F	G	H	I
						Column1		
						Monthly Average		
Row Label	Total Revenue	Total Orders	AVGRATING		Column		SEASONAL INDEX	
2023	3901123,83	4047	3,9058562		AOV		Seasonal Index	Above/Below Avg
January	286694,32	301	3,910299			325093,6525	0,881882245	Below average
February	338784,92	334	3,9461078		952,4728		1,042114841	Above average
March	321716,01	344	3,9127907		1014,326		0,989610248	Below average
April	257332,58	281	3,9359431		935,221		0,791564455	Below average
May	351902,03	355	3,9098592		915,7743		1,082463553	Above average
June	324356,96	349	3,8653295		991,2733		0,997733907	Below average
July	346618,03	342	3,8684211		929,3896		1,066209775	Above average
August	391505,89	371	3,9326146		1013,503		1,204286479	Above average
September	262602,86	306	3,8856209		1055,272		0,80777603	Below average
October	352286,61	361	3,8947368		858,1793		1,083646535	Above average
November	282606,1	336	3,952381		975,8632		0,86930673	Below average
December	384717,52	367	3,8637602		841,0896		1,183405204	Above average
2024	1014420,74	953	3,8898216		1048,277			
January	396874,92	358	3,8156425			338140,2467	1,17369915	Above average
February	342913,06	307	3,8794788		1108,589		1,014114893	Above average
March	274632,76	288	3,9930556		1116,981		0,812185957	Below average
Grand Total	4915544,57	5000	3,9028		953,586			
Day-of-Week analysis								
Row Label	Count of Order_ID	Sum of Total_Amount						
Sunday	705	666857,12						
Monday	738	817399,03						
Tuesday	738	768221,72						
Wednesday	667	670605,98						
Thursday	692	663871,24						
Friday	715	669175,46						
Saturday	745	659414,02						
Grand Total	5000	4915544,57						

Key Finding

The pivot tables provides a comprehensive view of business performance over 2023 and the first quarter of 2024. Revenue has grown steadily, with total orders increasing and a consistent average rating of around 3.9, reflecting moderate customer satisfaction. Monthly performance shows seasonal variations, with strong months like August and December and weaker periods like April and September. The day-of-week analysis reveals that Monday

and Tuesday drive the most revenue, while weekends are quieter. The average order value peaks during festive months, indicating high spending periods.

Business Implication

From these findings, several business implications emerge. The company is in a strong financial position, but it should use peak months to maximize premium offerings and focus on marketing strategies to lift weaker periods. Improving customer experience is essential to push ratings above 4.2, which could be achieved through better service training. Additionally, weekend promotional efforts may help balance demand across the week. Lastly, the business should leverage high average order values in festive months by offering bundles or upselling. Overall, these steps will sustain growth, improve customer satisfaction, and ensure the company remains competitive in the long run.

4.3 Customer and Demographic Analysis (Task 2.3)

Column Labels									
Female					Male				
Row Label	Average of Total_Am	Average of Customer_R	Average of Total_Am	Average of Customer_R	Other	Average of Total_Am	Average of Customer_R	Total Average o	Total Average of Customer_Rating
18-50	989,2970357	3,89442876	904,356958	3,923598553	830,3694286	779,5823333	3,8	984,4584766	3,902176302
51+	1027,879467	3,995555555	913,1443776	3,996809997				989,4965954	3,909090909
Grand Total	992,7806019	3,887239165	977,8351951	3,921149897	828,8576712		3,821917808	983,108914	3,9028

Age Group x Gender cross-analysis									

Product_Category per group									
Product_Category	Sports								
Row Label	Count of Order_ID								
18-50	327								
Female	327								
51+	30								
Female	30								
Grand Total	357								

1 City revenue rankings									
Row Labels	Total Revenue								
Istanbul	1334122,56								
Ankara	657535,82								
Izmir	567534,67								
Bursa	459076,31								
Adana	427059,63								
Konya	344374,94								
Gaziantep	319408,41								
Antalya	315549,89								
Kagseri	312303,91								
Eskisehir	178378,43								
Grand Total	4915544,57								

2 City revenue rankings									
Row Labels	AVERAGE TOTAL AMOUNT								
Kagseri	1215,190311								
Adana	1129,787381								
Konya	1088,249022								
Istanbul	1039,036262								
Izmir	945,891167								
Bursa	925,5570756								
Gaziantep	912,3450143								
Ankara	894,6065578								
Eskisehir	862,2782381								
Antalya	843,7162834								
Grand Total	983,108914								

Returning vs. New Customer									
Row Label	Average of Total_Am	Count of Payment_Met	Average of Delivers_Tin	Average of Customer_Rating					
TRUE	985,3601472	2990	6,459197324	3,897324415					
Grand Total	985,3601472	2990	6,459197324	3,897324415					

Customer preferred Payment Method									
Is_Returning	Payment_Method	Count							
FALSE	Credit Card	825							
TRUE	Debit Card	777							

RFM Segment profile									
Row Label	Count of Customer_ID	Sum of Total_Amount	Average of Customer_R	Average of Delivers_Time_Days					
(blank)	5000	4915544,57	3,9028	6,497					
Grand Total	5000	4915544,57	3,9028	6,497					

Key Finding

The age and gender analysis shows that customers aged 18–50 dominate sales, with average spending around R984 and ratings near 3.9. The 51+ group spends slightly more per order but represents a smaller share, making them valuable yet underutilized.

Business Implication

Marketing should continue targeting the 18–50 group while creating tailored campaigns for the 51+ segment to increase participation. Gender parity in spending suggests balanced appeal, but personalized promotions could further enhance engagement. Regional strategies should focus on maximizing Istanbul's volume while leveraging Kayseri and Adana's higher-value transactions with premium offerings. This reduces reliance on one city and diversifies revenue streams. Expanding sports-related product lines or introducing complementary categories like fitness accessories could strengthen the company's position among younger customers. Loyalty programs, personalized offers, and improved delivery times can enhance retention among returning customers. Payment method insights can guide partnerships with banks or platforms to streamline transactions.

Overall, the company should prioritize regional diversification, customer retention, and product expansion, while improving delivery efficiency and customer experience to sustain growth and competitiveness.

4.4 Digital Behaviour and Device Analysis (Task 2.4)

Key Finding

The device type comparison shows that mobile users generate the highest revenue (R2.81M) and have the largest order count, while desktops and tablets contribute less overall. Average ratings are similar across devices, but mobile slightly leads in engagement.

Engagement tier analysis reveals that medium-engagement customers spend the most (R1,011 average) and rate satisfaction at 3.9, while low-engagement customers spend far less (R653) and rate lower at 3.56. High-engagement customers spend less per order but give slightly higher ratings.

Payment method analysis shows credit cards dominate across all devices, with 2,012 transactions, followed by debit cards and digital wallets. Bank transfers and cash on delivery are less common, indicating a preference for digital and card-based payments.

Page view analysis indicates that customers who view 1–3 pages spend the most (R1,060 average), while those viewing 7–9 pages spend slightly less (R1,022). This suggests that quick, efficient browsing leads to higher-value purchases compared to prolonged navigation.

Business Implication

The dominance of mobile users highlights the need to prioritize mobile optimization, ensuring smooth app and site experiences to sustain revenue growth. Desktop and tablet experiences should still be improved to capture secondary segments.

Medium-engagement customers are the most profitable, so strategies should focus on keeping them active through personalized offers and loyalty programs. Low-engagement customers need targeted campaigns to increase both spending and satisfaction.

Credit card and digital wallet usage suggests that secure, seamless digital payment options are critical. Partnerships with banks and wallet providers could enhance convenience and encourage repeat purchases.

The page view findings imply that simplifying navigation and reducing friction in the purchase journey can increase order value. Streamlined product pages, faster checkout, and personalized recommendations could drive higher spending.

Device Type Comparison

Row Labels	Count of Total_Amount	Average of Total_Amount	Average of Customer_Rating	Sum of Total_Amount		
Desktop	1711	964,4	14,92	8,9649328	3,912916423	1650103,78
Mobile	2795	1006	14,35	9,0035778	3,900536673	2810468,64
Tablet	494	921	14,63	8,9412955	3,880566802	454972,15
Grand Total	5000	983,1	14,57	8,9842	3,9028	4915544,57

Engagement_Tier Performance

Row Labels	Average of Total_Amount	Average of Customer_Rating
High	929,304396	3,918
Low	653,438125	3,563
Medium	1011,429555	3,901
Grand Total	983,108914	3,903

Payment_Method × Device_Type cross-tabulation

Count of Order	Column Labels			
Row Labels	Desktop	Mobile	Tablet	Grand Total
Bank Transfer	153	307	50	510
Cash on Delive	88	129	31	248
Credit Card	678	1116	218	2012
Debit Card	439	713	113	1265
Digital Wallet	353	530	82	965
Grand Total	1711	2795	494	5000

Pages_Viewed bins analysis

Row Labels	Average of Total_Amount
7-9	1022,974573
4-6	980,1865217
1-3	1060,44806
10+	941,6882353
Grand Total	983,108914

4.5. Delivery Performance and Customer Satisfaction (Task 2.5)

Product Category delivery profile		
Row Labels	Average of Delivery_Time_Da	Average of Customer_Rating
Beauty	6,367149758	3,90821256
Electronics	6,386217949	3,919871795
Toys	6,418032787	3,883606557
Fashion	6,454983923	3,876205788
Books	6,465909091	3,967532468
Home & Garder	6,533011272	3,938808374
Food	6,625201939	3,830371567
Sports	6,709145427	3,898050975
Grand Total	6,497	3,9028

Delivery speed brackets		
Row Labels	Average of Customer_Rating	Count of Is_Returning_Customer
Fast (1-3 days)	3,950361944	967
Slow (8+ days)	3,896185116	1599
Standard (4-7 c	3,888249795	2434
Grand Total	3,9028	5000

Discount impact on satisfaction		
Row Labels	Average of Customer_Rating	Average of Delivery_Time_Days
FALSE	3,889464594	6,503742084
TRUE	3,933158585	6,481651376
Grand Total	3,9028	6,497

Interpretation

The product category delivery profile shows that delivery times average around 6.5 days across categories, with books and home & garden achieving slightly higher ratings (3.96 and 3.93) compared to food and fashion, which score lower. This indicates that while delivery speed is consistent, customer satisfaction varies by category, with some products perceived more positively.

Delivery speed analysis reveals that fast deliveries (1–3 days) achieve the highest ratings at 3.95, while slow deliveries (8+ days) drop to 3.89. Standard deliveries (4–7 days) dominate in volume but achieve only moderate ratings, showing that speed directly influences satisfaction and return behaviour.

Discount impact analysis shows that orders with discounts have slightly higher ratings (3.93) compared to non-discounted orders (3.88), despite similar delivery times. This suggests that discounts positively influence customer perception and satisfaction, even when operational performance remains unchanged.

Business Implications

The company should prioritize improving delivery speed, as faster fulfilment clearly boosts satisfaction and encourages repeat customers. Product categories with lower ratings, such as food and fashion, need quality improvements or better customer communication to raise satisfaction levels. Discounts are effective in enhancing customer perception, so strategic use of promotions can strengthen loyalty, but they should be balanced to avoid margin erosion. Overall, focusing on logistics efficiency, targeted category improvements, and smart discount strategies will enhance customer experience and drive sustainable growth.

Section 5: Dashboard

5.1 Dashboard Overview

The dashboard was developed in Microsoft Excel using PivotTables, PivotCharts, and Slicers. It delivers a single-screen executive view of the most critical business metrics, designed specifically for the COO to self-serve insights quickly and effectively, without requiring analyst support.



5.2 Dashboard Visuals — Description & Business Question

The full-page sales dashboard provides a comprehensive view of business performance across financial, operational, and customer dimensions. At the top, the KPIs summarize overall health with total revenue of R4.9 million, 5,000 orders, an average order value of R983, a satisfaction rating of 3.9, and delivery times averaging 6.5 days, while discounts account for 30.5% of sales. The monthly revenue trend highlights seasonal fluctuations, showing which months drive peaks and which underperform. Device analysis reveals mobile as the dominant sales channel, while desktop and tablet contribute less. Revenue by category shows electronics leading, followed by home & garden and fashion, while city analysis places Istanbul as the top contributor, with Kayseri and Adana delivering higher-value transactions despite smaller volumes. Delivery speed directly impacts satisfaction, with faster fulfilment linked to higher ratings, and category satisfaction shows home & garden and books outperforming food and fashion. Payment method distribution indicates a reliance on cash on delivery and credit cards, though digital wallets are also significant. Returning versus new customer analysis confirms loyalty as a major revenue driver, while day-of-week analysis shows Mondays and Tuesdays as peak sales days. Finally, browsing behaviour reveals that customers who view fewer pages tend to spend more, suggesting that streamlined navigation supports higher-value purchases. Altogether, the dashboard

answers critical business questions about what drives revenue, who the customers are, how they buy, and which operational factors most influence satisfaction and loyalty.

5.3 Slicer Logic

When explaining the slicer logic in the dashboard, the key point is that slicers act as interactive filters, allowing users to dynamically segment the data by dimensions such as gender, city, product category, order month, order year, and device type. Each slicer is connected to the underlying dataset so that when a selection is made, all visuals update simultaneously to reflect only the chosen subset. This ensures that comparisons are contextualized for example, selecting “Mobile” in the device slicer immediately shows how mobile users differ in revenue, ratings, and payment preferences compared to desktop or tablet users. Similarly, filtering by city or product category allows managers to drill into specific markets or product lines without altering the overall structure of the dashboard.

The key design decisions were made to balance clarity, usability, and insight. KPIs were placed at the top to provide an immediate snapshot of performance, while charts were organized logically: trends over time (monthly revenue) on the left, categorical breakdowns (device, product, city) in the centre, and behavioural or satisfaction metrics (delivery speed, ratings, payment methods, returning vs. new customers) on the right. Pie charts were used for proportions like payment methods and customer splits, while bar charts were chosen for categorical comparisons to make differences visually clear. The inclusion of slicers ensures flexibility, allowing the same dashboard to serve multiple analytical needs without overwhelming the user with separate reports. Altogether, these design choices create a dashboard that is both visually intuitive and strategically powerful, enabling quick decision-making while still supporting deeper analysis.

Section 6: Key Insights Summary

Executive Summary for the COO

The latest analysis of our sales and customer data highlights five critical insights that directly impact both revenue growth and operational efficiency.

1. **Strong overall financial performance.** The business generated R4.9 million in total revenue across 5,000 orders, with an average order value of R983. This demonstrates healthy demand and consistent sales volume, positioning the company on solid financial footing.
2. **Mobile dominance** in sales channels. Mobile devices accounted for the largest share of revenue, contributing R2.81 million compared to desktops (R1.65 million) and tablets (R455,000). This confirms that mobile optimization is essential, as the majority of customers are engaging and purchasing through mobile platforms.
3. **Geographic concentration** with untapped potential. Istanbul leads with R1.33 million in revenue, followed by Ankara and Izmir. However, smaller cities like Kayseri show the highest average order value at R1,215 per transaction, suggesting that premium offerings and localized marketing could unlock significant growth outside the main hubs.
4. **Delivery speed directly impacts satisfaction.** Fast deliveries (1–3 days) achieve the highest customer ratings at 3.95, while slow deliveries (8+ days) drop to 3.89. With average delivery times currently at 6.5 days, improving logistics could meaningfully raise satisfaction and loyalty.
5. **Discounts and loyalty drive perception.** Orders with discounts achieve slightly higher ratings (3.93 vs. 3.88 without discounts), showing that promotions positively influence customer sentiment. Returning customers already account for a significant share of revenue, with 2,390 repeat transactions, underscoring the importance of retention strategies.

Strategic Implications

These findings suggest clear priorities for the business. First, logistics improvements to reduce delivery times will directly enhance satisfaction and repeat purchases. Second, mobile platforms must remain the primary focus for optimization, given their revenue dominance. Third, geographic diversification into high-value smaller cities offers untapped potential. Fourth, targeted discount strategies can strengthen customer perception without eroding margins. Finally, loyalty programs should be expanded to build on the strong base of returning customers. Together, these actions will sustain growth, improve competitiveness, and ensure long-term profitability.

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